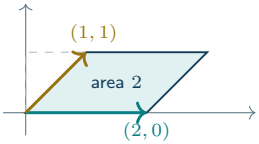
LINEAR ALGEBRA LADDER 3 OF 9

Matrices as transformations

A matrix moves the plane, its columns are where the basis vectors land, and its determinant is the area scale.

1 / THE RULE AND THE PICTURE

The columns of A are the images of $(1,0)$ and $(0,1)$, and everything else follows by combination



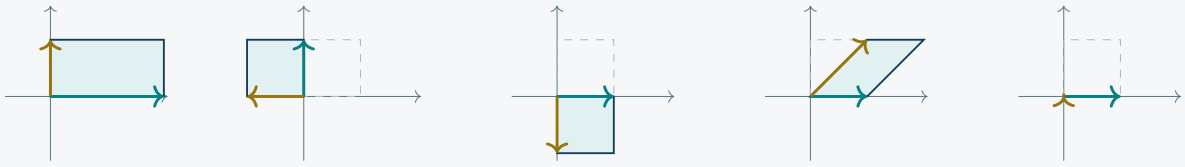
For $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ the unit square is carried to a parallelogram, because $A(1,0) = (2,0)$ and $A(0,1) = (1,1)$ are its two columns. A general point follows from linearity:

$$A \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 2 \\ 0 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The area of the parallelogram is 2, which is $\det A = 2 \cdot 1 - 1 \cdot 0$.

2 / A GALLERY OF 2×2 MAPS

Stretching, turning, flipping, shearing and flattening are five different matrices



stretch	rotation $90°$	reflection	shear	projection
$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

In each picture the dashed unit square becomes the shaded shape, with $(1,0)$ in teal and $(0,1)$ in gold. The determinants are $2,\ 1,\ -1,\ 1,\ 0$: the reflection reverses orientation and the projection flattens the square to a segment of area zero.

3 / COMPOSITION / PRODUCTS OF MATRICES

Doing one map after another multiplies the matrices, and the order matters



SR: rotate, then shear



RS: shear, then rotate

With the shear $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and the rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$,

$$SR = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad RS = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix},$$

so rotating first sends $(1,0)$ to $(1,1)$, while shearing first sends it to $(0,1)$. The determinants agree, $\det SR = \det RS = 1$, since $\det(AB) = \det A \det B$.

4 / SUMMARY / WHAT THE DETERMINANT SAYS

The determinant is the signed area scale, and zero means the map flattens the plane

map	$\det$	effect on area	inverse
stretch $\operatorname{diag}(2,1)$	2	doubles areas	invertible
rotation, shear	1	keeps areas	invertible
reflection	-1	keeps areas, flips orientation	invertible
projection	0	flattens to a line	not invertible

An invertible matrix can be undone, and the inverse sheet shows how, while a matrix with $\det = 0$ loses a direction for ever.

ASIDE / ROWS AND COLUMNS

Multiplying A by a vector takes combinations of the columns, while each entry of Ax is a row dotted with x.

LINKS TO OTHER SHEETS

The area scale is developed in *The determinant in its many guises*; the previous sheet is *Dot products, length and projection* and the next is *Inverses and elementary matrices*.

Convention. A acts on column vectors; $(1,0)$ and $(0,1)$ are the standard basis; $\det$ is the signed area scale factor.